

SENTINEL

**A Graph-Powered Forensic Intelligence Platform for Precision
Public Safety and Strategic Analysis**

Presented to: City Police Department Leadership



Empowering Citizens | Enhancing Transparency | Building Trust

The Problem

Citizens lack access to real-time, local crime data — creating fear, mistrust, and uninformed decisions.



No Visibility

Residents don't know what crimes occur in their own neighborhoods.



Reactive, Not Proactive

People learn about incidents only through social media — often inaccurate.



Trust Gap

Lack of transparency erodes public confidence in law enforcement.

What is Sentinel (GSCIP)?

Sentinel is a graph-powered forensic platform that transforms raw incident data into strategic situational awareness. It enables real-time surge detection, infrastructure risk modeling, and automated reporting, chat assistant—modernizing urban safety through precision spatio-temporal analytics.



Interactive Crime Map



Crime Trend Dashboards



Crime Incident search



Open Data Source



Chat Assistant



Official Reports

Sentinel Platform Features



Situational Map

Real-time forensic heatmapping with dynamic beat/ward/district overlays and forensic looping.



Tactical Forensics

Case-level exploration with advanced multi-select filtering for precincts and specific crime categories.



Trend Analysis

Mathematical surge detection comparing rolling 7-Day performance against a 30-Day strategic baseline.



Strategic Reports

Automated generation of production-ready multi-district forensic PDF portfolios for leadership briefings.

Sentinel Platform Features



AI Chat Assistant

A graph-powered natural language engine that allows investigators to query complex crime datasets instantly. Get rapid tactical answers (e.g., 'What are the top 3 crime types near hospitals in District-01?') without manual data sorting.



Integrated Infrastructure Locators

Holistic situational awareness by overlaying critical city assets onto forensic crime hotspots:

- Police Stations: Real-time proximity analysis for rapid dispatch and tactical coordination.
- Schools & Hospitals: Vulnerability mapping to prioritize protection and patrol density around sensitive public zones.
- Fire Stations: Emergency response overlap analysis in high-risk crime corridors to ensure safe and rapid access.

How It Works



Police log incident

Officers file case reports in the CPD system as usual — no extra steps.



Data is processed

Records are reviewed, classified by FBI/NIBRS codes and anonymized to block level.



Sentinel updates

The platform refreshes daily, keeping data current with a 7-day review buffer.



Public explores

Citizens search their neighborhood, view trends, and download open datasets.

Benefits for the Department



For the Police Department

- **Strategic Resource Allocation:** Deploy units based on forensic density peaks, ensuring maximum impact in high-risk zones.
- **Proactive Surge Detection:** Identify emerging crime trends(7-Day vs 30-Day) before they escalate into major city incidents.
- **Automated Command Briefings:** Generate professional, clinical PDF portfolios for leadership in seconds, eliminating manual data prep.
- **Advanced Forensic Search:** Query complex datasets using the AI Chat Assistant for instant situational answers.



For the Public

- **Real-Time Situational Awareness:** Understand localized safety trends through high-fidelity, interactive forensic heatmaps.
- **Infrastructure-Relative Safety:** Proactively assess risk proximity to critical assets like Schools, Hospitals, and Parks.
- **Enhanced Transparency:** Automated reporting ensures the public has access to the same forensic data used strategic policing.
- **Conversational Intelligence:** Use the AI Assistant to ask safety questions about specific neighborhoods and get instant, data-backed answers.

Privacy & Data Security

Block-Level Addresses Only



Victim and precise location data is never exposed. All incidents shown at street-block level only.

7-Day Review Buffer



Data is published with a 7-day delay, allowing time for review and reclassification before public display.

No Personal Identifiers



Names, ages, and identities of victims or suspects are never included in public-facing data.

Department-Controlled Data



The Police Department retains full control over what is published and when. A data review process governs all releases.



Let's Build a Safer, More Transparent City

Sentinel turns your department's existing data into a powerful public trust tool.
No new data collection. No operational disruption. Maximum community impact.

- Approve a pilot data-sharing agreement
- Assign a data liaison from the department
- Set a 6-month go-live target

Together, we protect and inform. | citymap@citypolice.gov | www.citymap.gov